

AI EMPLOYEE PLATFORM

Master Product, Technical & Implementation Specification

Outpero Parity → Differentiation → Working MVP

Prepared as the single source of truth for Cursor-driven implementation.

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1. Executive Summary

We are building a managed AI Employee platform for Indian businesses. The first objective is not to create an unnecessarily different product. It is to reach strong parity with the capabilities Outpero publicly presents, then add a simpler and more powerful Business Brain that keeps current business facts separate from conversational AI behaviour.

Outpero publicly presents AI voice agents for inbound/outbound calling, lead qualification, appointment booking, follow-up automation, WhatsApp, CRM/webhook/Pabbly integrations, Google Calendar booking, transcription, summaries, Telugu/English switching, bulk campaigns, live monitoring, and Indian telecom infrastructure.

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Our differentiator is that business owners should not have to understand how to train or configure AI. An internal AI Trainer prepares and monitors the employee, while the business owner manages business facts such as prices, stock, offers, FAQs and documents. This distinction is central to the product.

Important: Outpero's public site and product pages provide evidence of capabilities, but they do not expose every internal implementation detail. This document therefore distinguishes public Outpero capabilities from our proposed architecture rather than claiming knowledge of Outpero's private code or model stack.

2. Outpero Baseline — What We Need to Match

The following is the baseline to target before adding our differentiation. It is based on Outpero's current public website/product information researched for this document.

Capability	Outpero public evidence	Our priority
Inbound AI voice	24/7 answering, routing and FAQs are presented	Must Match
Outbound AI voice	Proactive calling and lead campaigns	Must Match
Lead qualification	Structured data captured from calls and AI lead qualification	Must Match
Appointment booking	Google Calendar booking and appointment automation	Must Match
Follow-ups	WhatsApp/SMS/follow-up sequences	Must Match
WhatsApp	Confirmations and follow-ups; WhatsApp automation	Must Match
CRM/webhooks	CRM, webhook and Pabbly integrations	Must Match
Call transcription	Automatic transcription	Must Match
Call summaries	Automatic summaries/outcomes	Must Match
Telugu + English switching	Native Telugu and switching to English mid-call	Must Match
Bulk campaigns	CSV campaigns and concurrent calling	Must Match
Call monitoring	Live monitoring and call visibility	Must Match
Indian +91 number	Dedicated DLT-registered +91 number is advertised	Must Match
DNC/retries	DNC enforcement and retry behaviour are advertised	Must Match
Low latency	Under-800ms response claim using voice infrastructure/models	Target
Business setup	Plain-English setup is advertised	Match UX, improve control
Pricing/usage	Per-minute usage, credits and employee concept	Later/Business model decision

3. What We Should Improve

- Business Brain: structured, authoritative business data is separate from conversational instructions.
- Dynamic prices/stock/offers: changes can take effect at a scheduled time without retraining the model.
- Internal AI Trainer: business owners do not need to train the AI themselves.
- Knowledge-gap workflow: failed questions become visible, actionable training/data tasks.
- Test Lab: every important business workflow can be tested before deployment.
- AI versioning: conversational behaviour changes are versioned and approved.
- Clear separation of business facts vs AI behaviour.
- Business-owner UI stays simple; internal trainer UI contains the complexity.
- Provider adapters keep telephony, voice, WhatsApp and future integrations replaceable.

4. Product Model



5. User Roles

5.1 Business Owner

- View AI employee status and performance.
- View calls, transcripts, leads, WhatsApp and appointments.
- Manage approved business information.
- Upload documents/information.
- Edit prices, stock and offers.

- Schedule future business-data changes.

5.2 Internal AI Trainer

- Prepare Business Brain.
- Configure AI Employee behaviour.
- Create tests.
- Run tests.
- Review calls.
- Resolve knowledge gaps.
- Create AI versions.
- Approve/deploy/rollback.

5.3 Customer

- Receive or make a call.
- Speak naturally in supported languages.
- Ask multiple follow-up questions.
- Request WhatsApp details.
- Provide lead information.
- Book an appointment.

6. End-to-End Product Flow

```

Business signs up
↓
Business information collected
↓
Internal AI Trainer assigned
↓
Information prepared
↓
AI Employee configured
↓
Test Lab
↓
Approve
↓
Deploy
↓
AI calls / answers customers
↓
Customer asks questions
↓
AI retrieves current Business Brain data
↓
AI answers
↓
Customer requests WhatsApp details
↓
WhatsApp sends details
↓
Lead / Appointment / Follow-up
↓
Call saved + monitored
↓
Trainer reviews failures
↓
Improve → Test → New Version → Deploy

```

7. Core Conversation Example

Business data: iPhone 15 128GB = ₹15,000; Pixel 9 128GB = ₹20,000; inventory available.

Customer: “iPhone 15 price entha?”

AI: Retrieves current iPhone price and answers.

Customer: “Pixel 9?”

AI: Performs a new product lookup and answers the Pixel price.

Customer: “Which one is cheaper?”

AI: Compares current retrieved records.

Customer: “Pixel 9 stock lo undha?”

AI: Checks current inventory.

Customer: “WhatsApp lo details pampinchandi.”

AI: Calls the WhatsApp action, then confirms only after the provider accepts the send.

8. Dynamic Business Data — Key Differentiator

```
CURRENT
iPhone 15 = ₹15,000

OWNER
Schedule new price = ₹17,000
Effective = Tomorrow 2:00 PM

1:59 PM
AI lookup → ₹15,000

2:01 PM
AI lookup → ₹17,000

MODEL RETRAINING
Not required
```

The authoritative value lives in the Business Brain/database, not inside the LLM's weights or a static prompt.

9. Technical Architecture

```
Browser
↓
Next.js + TypeScript
↓
Application API
  ■■■ Supabase Auth
  ■■■ PostgreSQL
  ■■■ Supabase Storage
  ■■■ Background Worker
  ■■■ AI Orchestrator
  ■■■ Structured Business Data
  ■■■ Knowledge Search / pgvector
  ■■■ Action Tools
  ■■■ WhatsApp
  ■■■ Calendar
  ■■■ Lead/CRM
  ■■■ Human Transfer

Voice path:
Phone → Telephony → Secure Media Stream → Realtime Voice → AI Orchestrator → Phone
```

Supabase publicly supports Postgres, Auth and Storage, and pgvector can be used for vector/embedding search inside Postgres. [cite](#) [turn0search1](#) [turn0search2](#)

Twilio's Media Streams documentation describes bidirectional audio over WebSockets for real-time voice applications; provider webhook/signature validation must be implemented. [■cite■turn0search3■](#)

10. Proposed Technology Stack

Layer	Choice	Reason
Frontend	Next.js + React + TypeScript + Tailwind	One modern web application
Backend/API	Next.js server/API layer + TypeScript	Avoid unnecessary second backend language
Database	PostgreSQL via Supabase	Relational business data + auth/storage ecosystem
Vector search	pgvector	Keeps initial knowledge retrieval inside Postgres
Auth	Supabase Auth	Integrated authentication
Storage	Supabase Storage	Private business documents/media
Voice/telephony	Provider adapters; initial Twilio + current supported realtime voice provider	Replaceable integrations
WhatsApp	Approved WhatsApp Business API/provider adapter	Provider-dependent compliance and messaging
Validation	Zod	Runtime validation
Testing	Unit + integration + end-to-end	Protect core business workflows
Background jobs	Small worker/runtime	Scheduled changes and asynchronous processing

11. Business Brain

Structured Data

- Products and variants.
- Prices with effective dates.
- Inventory.
- Services.
- Offers.
- Availability.

Knowledge

- FAQs.
- PDF/DOCX/TXT documents.
- Business policies.
- Product descriptions.
- Website information.

Rules

- Escalation.
- Allowed actions.

- Business restrictions.
- Conversation constraints.

12. AI Orchestrator and Tools

```
search_products
get_current_price
get_inventory
search_knowledge
send_whatsapp
create_lead
update_lead
check_availability
book_appointment
transfer_to_human
```

Every tool must receive business context, validate inputs, enforce permissions, return structured output, and report explicit errors.

13. Voice Flow

```
Phone Call
↓
Telephony Provider
↓
Secure Media Stream
↓
Realtime Voice Session
↓
AI Orchestrator
↓
Business Brain / Tools
↓
Voice Response
↓
Customer
```

Outpero publicly advertises dedicated +91 numbers, DLT-compliant calling, Telugu/English switching, bulk calling, live monitoring and low-latency voice infrastructure. These are parity targets, not claims about our implementation yet.

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14. WhatsApp Core Flow

```
Customer says:
"WhatsApp lo details pampinchandi."
↓
AI identifies requested information
↓
Retrieve current Business Brain data
↓
Build approved message/content
↓
WhatsApp provider
↓
Provider acceptance/status
↓
Save message + provider ID + status
↓
AI confirms only if appropriate
```

Outpero publicly lists WhatsApp confirmations/follow-ups and WhatsApp Business automation among its offerings.

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15. Trainer Improvement Loop

```

LIVE CALL
↓
TRANSCRIPT + TOOL EVENTS
↓
EVALUATION
↓
FAILURE / KNOWLEDGE GAP
↓
AI TRAINER
↓
FIX DATA OR BEHAVIOUR
↓
TEST
↓
NEW AI VERSION
↓
APPROVE
↓
DEPLOY

```

16. Frontend Information Architecture

Business Owner	AI Trainer
Dashboard	Trainer Dashboard
AI Employee	Businesses
Calls	Business Setup
Leads	Business Data
WhatsApp	Knowledge
Appointments	AI Configuration
Business Information	Test Lab
Settings	Conversations
	Knowledge Gaps
	Versions
	Deployment

17. Key Screens

Business Dashboard

Show AI status, calls today, leads, WhatsApp sends, appointments, recent conversations and alerts.

Business Information

Products, services, offers, FAQs and documents. Business owners manage facts, not AI internals.

Trainer Test Lab

Input → expected → actual → source → tool calls → latency → pass/fail.

Knowledge Gap

Customer question → AI failure → reason → resolve → retest.

AI Versions

Draft → Testing → Approved → Live; keep rollback capability.

18. Core Database Model

Entity	Purpose
businesses	Tenant/business account
business_members	Business roles
ai_employees	AI employee identity/configuration
ai_versions	Versioned conversational behaviour
products / variants	Structured product facts
product_prices	Effective-dated prices
inventory	Current stock
services / offers	Business offerings
scheduled_changes	Future business-data changes
knowledge_documents / chunks	Uploaded knowledge + vectors
business_faqs / rules	Structured knowledge and rules
customers / leads	Customer and sales context
calls / call_messages	Conversation records
call_tool_events	Action trace
whatsapp_messages	WhatsApp delivery lifecycle
appointments	Booking lifecycle
ai_tests / ai_test_runs	Evaluation
knowledge_gaps	Failures requiring trainer attention
audit_logs	Important changes

19. Security and Data Rules

- Business-owned data must be tenant-isolated with RLS.
- Provider secrets stay server-side.
- Webhook signatures must be validated.
- Private documents must not be publicly accessible.
- Never expose secrets in frontend bundles.
- Never retrieve knowledge across businesses.
- Important business changes are audited.
- Scheduled jobs must be idempotent.
- Unknown information must never be fabricated.

20. Implementation Roadmap

Phase	Goal
1	Repository + environment + base UI
2	Database + Auth + RLS
3	Application shell/routes
4	Business onboarding
5	Business Brain structured data
6	File upload + knowledge
7	AI Trainer Console
8	AI Test Lab
9	AI Orchestrator + tools
10	Voice integration
11	WhatsApp
12	Leads + appointments
13	Business dashboard
14	Continuous improvement
15	AI versioning + deployment
16	Full testing + edge cases
17	UI polish/accessibility
18	Security review
19	Production deployment
20	Pilot with 3–5 businesses

21. Outpero Parity Checklist

Feature	Parity target	Our enhancement
Voice calling	Inbound/outbound, 24/7	Business-aware runtime retrieval
Telugu/English	Natural switching	Tanglish + broader Indian-language evaluation
Lead qualification	Structured call fields	Configurable test-backed extraction
Appointments	Calendar booking	Unified business action layer
WhatsApp	Follow-ups/confirmations	Voice → current details → WhatsApp
CRM/webhooks	Integrations	Provider adapters + audit trail
Transcription	Automatic	Conversation + tool trace
Summaries	Automatic	Trainer evaluation loop
Bulk campaigns	Concurrent calling	Controlled campaigns + observability
Monitoring	Dashboard visibility	Knowledge-gap + quality workflow

Feature	Parity target	Our enhancement
Business setup	Plain English	Internal Trainer + simple owner controls
Knowledge	Business instructions/data	Business Brain with authoritative structured facts
Prices/stock	Business-specific answers	Scheduled effective-date changes without retraining

22. MVP Acceptance Test

1. Business logs in.
2. AI Employee is live.
3. iPhone 15 = ■15,000.
4. Pixel 9 = ■20,000.
5. Customer receives a call.
6. Customer asks iPhone price → ■15,000.
7. Customer asks Pixel → correct Pixel price.
8. Customer asks which is cheaper → current comparison.
9. Customer asks stock → current inventory.
10. Customer asks for WhatsApp → details sent + status saved.
11. Lead is saved.
12. Appointment can be booked.
13. Transcript is saved.
14. Business sees the call.
15. Trainer inspects the conversation.
16. Trainer can resolve a knowledge gap.
17. Business schedules iPhone ■15,000 → ■17,000 at 2 PM.
18. Before 2 PM AI returns ■15,000.
19. After 2 PM AI returns ■17,000.
20. No model retraining occurs.
21. Trainer changes AI behaviour separately.
22. New AI version is tested before deployment.

23. Cursor Build Contract

- Read the complete specification before coding.
- Treat this PDF as the source of truth.
- Build in dependency order.
- Keep the app runnable after each phase.
- Do not invent unrelated features.
- Do not put API keys or secrets in code/docs.
- Use provider adapters for external systems.
- Business data is runtime truth; prompts are not the database.
- AI behaviour changes require versioning/testing/approval.
- Business data changes must work without retraining.
- Do not claim an external action succeeded without confirmed provider state.
- Do not expose technical AI concepts to business users unless explicitly required.
- Before production, run the complete end-to-end acceptance test.

24. Research Notes and Evidence

Outpero's public solutions page currently lists 19 standalone solutions including WhatsApp automation, lead qualification, appointment booking, follow-up sequences, inbound/outbound/custom voice agents, and other business automation solutions. ■cite■turn0search0■

Outpero's current pricing page publicly lists one AI employee for a 30-day hiring fee, a dedicated +91 line, bulk campaigns, Telugu/English switching, automatic transcription and summaries, CRM/webhook/Pabbly integrations, Google Calendar booking, WhatsApp confirmations/follow-ups, and multiple voice tiers. ■cite■turn0search1■

Outpero's deploy page publicly describes plain-English setup, lead-triggered calling, 24/7 inbound support, CSV mass campaigns, DNC/retry handling, live monitoring, DLT-compliant +91 numbers, and low-latency voice infrastructure. ■cite■turn0search3■

Outpero's industry pages show use cases such as win-back, feedback, reminders, website/Instagram enquiry callbacks, offers, catering/party booking confirmations, COD confirmation, abandoned-cart calls, and appointment reminders. ■cite■turn0search5■turn0search6■turn0search7■

These sources establish the public feature baseline. They do not establish Outpero's private source code, exact internal model training procedure, private API keys, or proprietary backend architecture. Those details should not be represented as confirmed facts.